MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM,
(POLARIZED, PLAIN FOIL), 85°C (VOLTAGE DERATED TO 125°C),
ESTABLISHED RELIABILITY,
STYLE CLR35

This amendment forms a part of Military Specification Sheet MIL-C-39006/3C, dated 10 January 1972, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Following preamble, add:

"CAUTIONARY NOTE: THESE CAPACITORS MAY EXPLODE IF SUBJECTED TO REVERSE VOLTAGES. WHEN FIRST TESTED FOR DC LEAKAGE OR AT OTHER HIGH POTENTIALS, EXERCISE CAUTION TO VERIFY CORRECT POLARITY MARKINGS."

"FAILURE RATE LEVEL 'L' INACTIVE FOR DESIGN AFTER 27 May 1976"

"CASE SIZE E1 INACTIVE FOR DESIGN AFTER 20 Oct 1978"

FIGURE 1 (dimension table): Add G1 case size

G1	.688 (17.48)	.188 (4.78)	.219 (5.56)	1.188 (30.18)

PAGES 2 AND 3

TABLE I, following part no. M39006/03-: Add "1/".

TABLE I, part number column: Add "-" following numbers 1349 through 1544 to denote optional random vibration part numbers.

TABLE I: Add the new G1 case size units

Rated voltage (85°C)	Cap.	Cap. tolerance	DC leakage (max) 25°C	DC leakage (max) 85°C 125°C	De-rated voltage (125°C)	Surge voltage (85°C)	Impedance (max)	Case size	Part no. M39006/03- 1/ Failure rate level (%/1,000 hr) L (2.0)	M (1.0)	P (0.1)	R (0.01)	S (0.001)
15	4.5	±20	1	2	10	17.2	495	G1	2700	2712-	2724-	2736-	2748-
25	3.0	±20	1	2	15	28.8	744	G1	2701	2713-	2725-	2737-	2749-
30	2.5	±20	2	5	20	34.5	893	G1	2702	2714-	2726-	2738-	2750-
50	1.5	±20	1	4	30	57.5	1487	G1	2703	2715-	2727-	2739-	2751-
75	1.0	±20	1	4	50	86.2	2232	G1	2704	2716-	2728-	2740-	2752-

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TABLE I (Continued): Add the new G1 case size units.

"

Rated voltage (85°C)	Cap.	Cap. tolerance	DC leakage (max) 25°C	DC leakage (max) 85°C 125°C	Derated voltage (125°C)	Surge voltage (85°C)	Impedance (max)	Case size	Part no. M39006/03- 1/ Failure rate level (%/1,000 hr) L (2.0)	M (1.0)	P (0.1)	R (0.01)	S (0.001)
100	0.8	±20	1	3.5	65	115.0	2790	G1	2705	2717-	2729-	2741-	2753-
150	0.5	±20	2	4	100	172.0	4454	G1	2706	2718-	2730-	2742-	2754-
200	0.35	±15	4	25	150	230.0	5577	G1	2707	2719-	2731-	2743-	2755-
250	0.3	±15	6	32	165	287.0	6600	G1	2708	2720-	2732-	2744-	2756-
300	0.27	±15	6	40	200	345.0	9000	G1	2709	2721-	2733-	2745-	2757-
375	0.22	±15	6	80	250	431.0	11800	G1	2710	2722-	2734-	2746-	2758-
450	0.15	±15	8	120	300	518.0	21400	G1	2711	2723-	2735-	2747-	2759-

"

Following Table I: Add "1/ Dash number for failure rate level M, P, R, and S parts will include the letter 'H' to indicate the optional random vibration requirement of '-' will be deleted."

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Following vibration requirement, add new Random vibration requirement:

"Random vibration (when applicable): Method 214 of MIL-STD-202, test condition II-E (18.7 g)."

Temperature cycling: Delete "Temperature cycling: Method 102 of MIL-STD-202, condition D" and substitute "Thermal shock: Method 107 of MIL-STD-202, condition A."

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Life at 85°C: Delete "240-hr (group B)" requirement in its entirety; also delete "10,000-hr (group C)" and substitute "10,000-hr (group B)".

Add the following new requirement under "Life at 125°C":

"10,000-hr (group B):

DCL (125°C) - 125% (max) of table I value.

DCL (25°C) - See table I.

ΔCap. - Within ±50% of initial measurement for E1 and G1 cases; within ±25% of initial measurement for all other cases.

DF - 40 (max) for E1 and G1 cases; 20% (max) for all other cases.

DWV - 2,000 V dc, min.

IR - 100 megohms, min."

Marking: Delete requirement in its entirety and substitute:

"Marking: In accordance with MIL-C-39006.

Polarity - Series of plus (+) symbols encircling positive end (see figure 1)."

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NOTE: The margin of this amendment is marked with an asterisk to indicate where a change (additions, modifications, corrections, deletions) from the previous amendment was made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - AR
Air Force - 17, 85, 99
DLA - ES

User activities:
Navy - AS, MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

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